

Automated Mapping of Lyrical Content to Life Themes: An Empirical and Computational Taxonomy

The systematic classification of popular music lyrics into recurring life themes represents a critical intersection of computational linguistics, music information retrieval (MIR), and music psychology. Historically, music has functioned as an emotional and narrative canvas, capturing personal introspection and broader socio-cultural transitions. Grounding a "life themes" axis in robust academic literature requires synthesizing longitudinal trends, structural and linguistic genre boundaries, and the psychological mechanics of musical preference. This report details the empirical foundations of lyrical content analysis, proposes a validated taxonomy of eight core life themes, explores the computational mathematics of semantic granularity, and outlines a multimodal roadmap for processing instrumental and ambiguous tracks.

Historical Evolution and Cross-Decade Trends in Lyrical Content

A rigorous taxonomic framework must account for the temporal dynamics of lyrical expression, as popular music consistently mirrors shifts in cultural values and psychological baselines. Longitudinal analyses of popular music lyrics over the past seven decades demonstrate systematic, statistically significant changes in both emotional valence and structural complexity.

Emotional Valence and the Rise of Negativity

Large-scale automatic sentiment analyses of the Billboard Hot 100 charts reveal a clear, decadal transition toward negative emotional tones. A primary study conducted by Napier and Shamir (2018) analyzed 6,150 Billboard songs spanning 1951 through 2016, demonstrating that markers of anger, disgust, fear, and sadness have increased significantly, whereas indicators of joy, confidence, and openness have steadily declined. This downward trend in average valence is highly consistent across Western popular music, indicating that songwriters increasingly utilize music to process psychological distress, social alienation, and systemic conflict rather than to express collective celebration or romantic idealism.

The Individualistic Shift and Linguistic Self-Focus

This emotional trajectory is closely coupled with a documented shift in linguistic self-focus. Using computerized text analysis on top Billboard singles from 1980 to 2007, DeWall et al. (2011) identified a persistent increase in first-person singular pronouns (such as "I," "me," and "my") alongside a corresponding decrease in first-person plural pronouns (such as "we" and "us"). This rising self-focus is strongly correlated with an uptick in antisocial behavioral markers and negative emotions, suggesting that contemporary songwriting has grown increasingly self-referential, introspective, and individualistic.

Structural Simplification and the Rise of Explicitness

Simultaneously, the structural complexity of popular lyrics has evolved. Over the last five decades, popular music across genres has become systematically simpler and more repetitive. Lexical complexity—measured via vocabulary richness and readability indices—has decreased, while the repetition of structural choruses and hooks has increased. Furthermore, from 1990 onwards, there has been a dramatic surge in the prevalence of profane, sexually explicit, and aggressive language in mainstream charts. This rise in explicit content, validated by deep learning language models applied to longitudinal Billboard datasets, reflects a broader relaxation of institutional censorship and a shift in societal standards regarding language use.

Cross-Genre Thematic Profiles and Narrativity Dynamics

While decades shape the overall emotional climate of popular music, genre remains the primary structural boundary that dictates thematic content, verbal complexity, and narrative style. Computational studies indicate that popular genres display distinct, highly specialized thematic clusters.

Narrativity and Storytelling Patterns

To map how stories are constructed in contemporary music, a critical distinction must be made between narrative lyrics—which construct a coherent chronological sequence of events—and impressionistic lyrics, which evoke non-narrative emotional states or sensory impressions. A study conducted at the University of California, Berkeley, utilized a machine learning algorithm to analyze narrative storytelling elements in over 5,000 Billboard Hot 100 songs from 1960 to 2024.

The findings challenged the common assumption that storytelling in popular music peaked during the folk-revival era of the 1960s with artists such as Bob Dylan, Joan Baez, and Simon & Garfunkel. Instead, the computational analysis revealed that narrativity has been on a steady uptick since the 1990s, driven primarily by the mainstream rise of hip-hop.

The linguistic structure of hip-hop is exceptionally verbose, averaging a significantly higher word length and unique vocabulary count than any other genre. This verbosity enables dense, highly sequential storytelling. For example, Ice Cube's *It Was a Good Day* was computationally ranked as the fifth most narrative song in the 60-year dataset, depicting a detailed, chronological "day in the life" of a protagonist navigating romance, community connection, systemic racism, and police surveillance in South Central Los Angeles.

This stands in stark contrast to non-narrative, impressionistic tracks like David Bowie's *Fame*, which utilizes repetitive, abstract declarations to evoke the psychological state of celebrity without establishing a sequential storyline.

Genre-Specific Thematic Environments Thematic differentiation across genres is highly pronounced, reflecting the distinct cultural, socioeconomic, and psychological environments from which these genres emerge.

- **Rock and Metal:** These genres feature low structural repetition and high lexical density. Thematically, they focus heavily on anti-conformist identity crises, institutional skepticism (such as "school as a prison" motifs), existential dread, and mortality.

- **Country:** This genre displays high narrative coherence and a strong attachment to tradition, community, and geographical place. Country lyrics consistently explore themes of family dynamics, adjusting with dignity to financial hardships, nostalgic reflection, and marital devotion.
- **Pop and R&B:** Pop and R&B are structurally characterized by high repetitiveness, optimized for acoustic appeal and memorability. Thematically, they concentrate on the interpersonal spectrum, ranging from celebratory romantic intimacy and physical seduction to the psychological fallout of betrayal and heartbreak.
- **Indie and Alternative:** Leaning away from mainstream pop structures, these genres prioritize abstract self-perception, emotional vulnerability, and existential inquiry.
- **Electronic and Dance:** These genres feature low narrative storytelling and high structural repetition, utilizing minimal lyrics to emphasize acoustic energy and physical movement over complex semantic arguments.

Empirical Dominance and Distribution of Lyrical Themes

To ground a computational taxonomy in empirical evidence, a mapping engine must prioritize themes that consistently dominate large-scale, peer-reviewed content-analysis studies. A core benchmark is the observational study conducted by Kwon et al. (2021), which analyzed 1,052 American and Latin songs across five major genres (Hip Hop/R&B, Pop, Country, Rock/Metal, and Latin) charting on the Billboard Hot 100 between 1998 and 2018.

Thematic Distribution and Cross-Genre Variations

The study identified 46 distinct lyrical themes, categorizing them into positive, neutral, and negative influencers based on their potential psychological impact on adolescent mental health. Table 1 synthesizes the empirical distribution of these dominant themes across genres as established by Kwon et al..

Lyrical Theme Category	Hip-Hop/R&B (n=208)	Pop (n=212)	Country (n=210)	Rock/Metal (n=210)	Latin (n=209)
Love & Relationships	30.29%	58.96%	50.00%	35.07%	57.89%
Sex & Sexual Innuendo	44.23%	16.98%	10.00%	8.53%	30.14%
Wealth, Fame & Vanity	42.31%	8.49%	0.48%	2.84%	0.96%
Alcohol Use	30.29%	11.79%	29.52%	5.69%	7.66%
Weapons & Violence	15.38%	2.36%	1.43%	6.16%	0.48%
Faith & Spirituality	0.00%	0.47%	11.90%	4.74%	0.48%
Betrayal & Relational Hurt	0.00%	0.94%	15.71%	8.06%	29.19%
Loneliness &	1.44%	1.42%	10.00%	8.53%	2.39%

Lyrical Theme Category	Hip-Hop/R&B (n=208)	Pop (n=212)	Country (n=210)	Rock/Metal (n=210)	Latin (n=209)
Isolation					
Resilience & Coping	0.96%	2.36%	5.71%	9.95%	0.96%
Suicide & Mortality	3.37%	0.00%	0.00%	17.06%	0.00%

Statistical Insights and Co-Occurrence Dynamics

Kwon et al. reported a significant 180\% increase in the percentage of songs containing negative themes across all charting music between 1998 and 2018 (P <.001), while positive and neutral themes remained statistically static (P =.54 and P =.26, respectively). The distribution of these themes is highly unequal across genre boundaries. Hip-Hop/R&B demonstrated a significantly higher density of negative themes (62.5\%, P <.001) compared to any other genre, driven by high concentrations of references to drugs, weapons, and material wealth. Conversely, Hip-Hop/R&B featured a significantly lower frequency of positive themes compared to Pop (P =.03), Country (P =.003), and Rock/Metal (P =.002). Furthermore, the data reveals critical co-occurrences. For example, substance use (such as alcohol and drugs) heavily overlaps with themes of relational betrayal, loneliness, and mental health struggles. This indicates that substance use in popular lyrics frequently transitions from a celebratory marker of high status to a chemical coping mechanism for processing emotional pain. In psychology, these tracks are recognized as "coping songs," selected by listeners to regulate difficult emotional states, vent anxiety, or find solace through shared experience.

Proposed Lyrical Life Theme Taxonomy

To map song lyrics and artist content effectively, this report proposes an academically grounded, eight-category taxonomy. This framework consolidates overlapping academic models into distinct, mutually exclusive axes optimized for computational classification.

1. Interpersonal Devotion and Bonds

- **Definition:** The celebration, romanticization, and preservation of deep emotional, romantic, or familial connections. This category focuses on attachment, security, and the idealization of another person.
- **Common Lyrical Markers:** High concentration of second-person pronouns ("you," "your"), words of physical proximity and attachment ("hold," "stay," "belong," "devoted"), and idealizing imagery (stars, light, dreams, forever).
- **Example Artists or Tracks:** Al Green (*Let's Stay Together*), Taylor Swift (*Lover*), Shania Twain (*You're Still the One*).

2. Heartbreak, Infidelity, and Relational Loss

- **Definition:** The emotional distress, mourning, anger, and psychological fallout resulting from the dissolution of a romantic relationship, unrequited love, or betrayal.
- **Common Lyrical Markers:** High frequency of sad and distress-related words ("cry,"

"tear," "cold," "empty," "broken"), past-tense memories of intimacy, and terms of betrayal or infidelity ("lie," "cheat," "fool," "wrong").

- **Example Artists or Tracks:** Adele (*Someone Like You*), Olivia Rodrigo (*Drivers License*), Amy Winehouse (*Back to Black*).

3. Adversity, Mental Health, and Resilience

- **Definition:** The internal and external battle against physical or psychological pain, depression, self-doubt, trauma, or systemic struggle, paired with the pursuit of resilience.
- **Common Lyrical Markers:** Vocabulary associated with psychological distress ("darkness," "pain," "walls," "tiredness," "chaos"), self-harm or suicidal ideation, and themes of endurance ("fight," "survive," "stronger," "scars").
- **Example Artists or Tracks:** Linkin Park (*In the End*), Eminem (*Lose Yourself*), Kelly Clarkson (*Stronger*).

4. Identity, Anti-Conformity, and Autonomy

- **Definition:** The pursuit of self-actualization, self-acceptance, the rejection of social pressure and institutional conformity, and the definition of personal values.
- **Common Lyrical Markers:** High frequency of first-person singular pronouns ("I," "me," "my"), assertions of authenticity ("true," "free," "born," "mirror"), and systemic rebellion ("chains," "cage," "prison," "rules").
- **Example Artists or Tracks:** Dead Kennedys (*Life Sentence*), Lady Gaga (*Born This Way*), Weezer (*The Greatest Man That Ever Lived*).

5. Status, Ambition, and Material Dominance

- **Definition:** The pursuit and celebration of financial wealth, social fame, personal influence, and competitive dominance over peers or rivals.
- **Common Lyrical Markers:** Concrete nouns representing wealth and luxury ("money," "gold," "diamonds," "clown," "crown"), assertions of victory and power ("win," "rule," "hustle," "boss"), and highly assertive self-aggrandizement.
- **Example Artists or Tracks:** Jay-Z (*Dirt Off Your Shoulder*), David Bowie (*Fame*), Drake (*Started from the Bottom*).

6. Hedonism, Party, and Substance Escapism

- **Definition:** The pursuit of sensory pleasure, dance, sexual freedom, and chemical escapism as a means of celebration or temporary release from life concerns.
- **Common Lyrical Markers:** Explicit references to substances ("pour," "drink," "smoke," "pills," "booze"), terms of kinetic movement and space ("club," "dance," "floor," "night"), and temporal urgency ("tonight," "never stop").
- **Example Artists or Tracks:** LMFAO (*Party Rock Anthem*), The Weeknd (*Can't Feel My Face*), Rihanna (*Cheers (Drink to That)*).

7. Place, Heritage, and Nostalgia

- **Definition:** Emotional grounding in a specific geographic location, childhood home,

cultural heritage, or the comforting memories of the past and family ties.

- **Common Lyrical Markers:** Spatial and travel nouns ("hometown," "road," "train," "river"), temporal markers of memory ("yesterday," "remember," "years ago"), and familial terms ("mother," "father," "home").
- **Example Artists or Tracks:** John Denver (*Take Me Home, Country Roads*), Bruce Springsteen (*Thunder Road*), Kacey Musgraves (*Merry Go 'Round*).

8. Existentialism, Mortality, and Spirituality

- **Definition:** Contemplation of the transience of life, the inevitability of death, cosmic meaning, religious faith, and the search for higher truth.
- **Common Lyrical Markers:** Metaphysical and religious vocabulary ("soul," "heaven," "grave," "shadow," "god," "pray," "ghost"), questions of cosmic purpose, and symbols of temporal transience.
- **Example Artists or Tracks:** Johnny Cash (*The Man Comes Around*), Hozier (*Take Me to Church*), Iron Maiden (*The Number of the Beast*).

Optimal Semantic Granularity and Hierarchical Architectures

When designing a computational mapping engine, determining the level of semantic granularity is a critical architecture choice. The system must balance the empirical reality of lyrics with user expectations.

The Failure of Flat, Single-Label Models

Applying a flat, single-label classification model to song lyrics is empirically weak and theoretically unsupported. Unlike expository prose, song lyrics are highly compressed, metaphorical, and context-dependent. A single song often traverses multiple semantic domains. For instance, a track may narrate a breakup by jumping between memories of early devotion (Interpersonal Devotion), feelings of loneliness (Adversity & Struggle), and final emotional recovery (Identity & Autonomy).

Linguistic experiments conducted by Papazoglou and Gaizauskas (2021) demonstrated that forcing a song into a single, highly specific category yielded poor classification performance because many adjacent categories share similar word distributions (e.g., distinguishing "heartbreak" from "unrequited longing").

Instead, the researchers demonstrated that a top-n classification approach (where $n \geq 3$) is optimal. This multi-label architecture allows the model to assign multiple, weighted themes to a single song, reflecting its narrative and emotional progression.

Hierarchical Topic Modeling and Regularization

To resolve the granularity puzzle, computational musicology utilizes Hierarchical Lifelong Topic Models (HLTM) and Hierarchical Latent Dirichlet Allocation (hLDA). These models organize themes into a tree structure, where higher-level nodes represent broad, abstract categories (e.g., "Interpersonal Dynamics") and lower-level child nodes represent granular sub-themes (e.g., "Romantic Devotion," "Heartbreak," "Infidelity").

However, standard hierarchical models frequently suffer from **Low Rationality** (where child topics are identical in granularity to their parents) and **Low Diversity** (where sibling topics become highly repetitive). To mitigate this, systems apply **Transport Plan Dependency (TPD)** constraints.

TPD acts as a mathematical regularizer during model training, constraining dependencies to ensure sparsity and balance. It forces a child topic close to its direct parent while pushing it away from other nodes, preventing semantic collapse.

This is paired with a **Context-Aware Disentangled Decoder (CDD)**, which decodes the documents using topics at each level of the hierarchy independently while incorporating a semantic bias from neighboring levels. This explicitly distributes different semantic granularities to different levels of the tree, ensuring that the model can differentiate broad life themes from specific emotional nuances.

Emotional Mapping and the Circumplex Model

Underlying these semantic hierarchies is Russell's Circumplex Model of Affect, which maps emotional states across two orthogonal dimensions: Valence (pleasantness to unpleasantness) and Arousal (energy to sleepiness). The MoodyLyrics dataset utilizes this dimensional framework to distribute song lyrics into four basic emotional quadrants :

By anchoring the lyrical life themes in these dimensions, a mapping tool can dynamically track the emotional intensity of a theme. For example, the *Heartbreak* category can be sub-divided into high-arousal states like "anger and vengeance" and low-arousal states like "grief and depression," aligning the taxonomy with established psychological metrics.

Engineering Roadmaps for Instrumental and Ambiguous Music

Purely text-based NLP systems fail when confronted with instrumental music, tracks with minimal lyrics (common in electronic dance music), or songs with highly metaphorical and ambiguous language. To resolve this "semantic gap," mapping engines must implement Multimodal Music Emotion Recognition (MMER).

Acoustic Feature Extraction and Mapping

In the absence of clear verbal content, the thematic and emotional weight of a track is carried by its sonic properties. These properties are extracted as quantitative descriptors across four primary perceptual dimensions. Table 2 maps these acoustic features to their corresponding thematic alignments in the proposed taxonomy.

Acoustic Dimension	Computational Feature Extraction Markers	Psychological Alignment	Core Thematic Alignment
Rhythm & Tempo	Onset Event Density , Beat Location agreement , Average Tempo (BPM).	Fast tempo and high density evoke high physiological arousal.	Status & Ambition , Hedonism & Party.
Dynamics & Energy	Root-Mean-Square (RMS) Energy , Attack	High dynamics and rapid attack phases	Adversity & Struggle , Status & Ambition.

Acoustic Dimension	Computational Feature Extraction Markers	Psychological Alignment	Core Thematic Alignment
	Slope , Volume Variance.	evoke anger, power, or dominance.	
Spectral Timbre	Mel-Frequency Cepstral Coefficients (MFCCs) , Spectral Centroid (Brightness).	Bright timbres denote happiness; dark timbres evoke sadness or existential dread.	Interpersonal Devotion (Bright) , Existentialism & Mortality (Dark).
Harmony & Tonality	Major/Minor Mode , Key Strength , Consonance ratio.	Major chords suggest positive valence; minor chords suggest negative valence.	Place & Nostalgia (Major) , Heartbreak & Loss (Minor).

Multimodal Fusion Architectures

To combine text embeddings and acoustic features, MMER systems employ multimodal deep learning architectures. Research demonstrates that multimodal systems significantly outperform single-channel classifiers, raising valence and arousal prediction accuracy. For example, a multimodal model integrating audio signal processing and visual video features achieved an emotional classification accuracy of 88.56\% (F_1 score of 0.88, \text{AUC} = 0.987). To achieve optimal results, mapping engines should avoid simple early concatenation or late voting in favor of **hybrid fusion utilizing cross-modal attention mechanisms**. At the heart of this approach is a self-attention mechanism that learns the complex, non-linear relationships between the text and audio branches.

If a song features highly ambiguous lyrics, the self-attention layer dynamically down-weights the text embedding and relies on the acoustic branch (which, for instance, may detect a slow tempo, legato articulation, minor mode, and soft dynamics) to classify the track under *Existentialism, Mortality, and Spirituality*. Conversely, if the lyrics are explicit and clear, the model prioritizes the semantic text branch.

Leveraging External Interpretive Corpora

To resolve highly metaphorical lyrics, mapping tools must leverage external interpretive corpora. Songwriters frequently break syntactic rules and deploy abstract metaphors that evade literal bag-of-words analyses.

By ingesting crowd-sourced annotations and interpretations (e.g., from Genius or Songfacts), the model can process standard natural-language explanations of specific song fragments. Linguistic evaluations show that incorporating reader interpretations significantly improves classification accuracy. Because interpretations are written in clear, expository prose, they act as an explanatory bridge, translating highly abstract lyrical figures of speech into concrete semantic concepts that the mapping engine can categorize with high confidence.

Synthesized Conclusions and System Architecture Recommendations

To construct a robust, state-of-the-art computational mapping engine that successfully maps song lyrics and artist content to recurring life themes, the following architectural choices are

recommended:

- **Implement the 8-Category Taxonomy:** Adopt the validated taxonomy (Interpersonal Devotion; Heartbreak & Loss; Adversity & Struggle; Identity & Autonomy; Status & Ambition; Hedonism & Party; Place & Nostalgia; Existentialism & Spirituality) to maximize empirical coverage and minimize training-label conflict.
- **Utilize a Multi-Label, Top-n Output Layer:** Avoid flat, single-label classifiers. Implement a multi-label output (n ≥ 3) to reflect the multi-faceted, narrative, and emotional progression of a single track.
- **Apply Hierarchical Topic Modeling with TPD Constraints:** Build a hierarchical tree (HLTM) to allow users to navigate from broad life themes down to granular sub-themes. Apply Transport Plan Dependency (TPD) constraints and a Context-Aware Disentangled Decoder (CDD) to prevent hierarchical collapse and ensure semantic diversity.
- **Deploy Hybrid Multimodal Fusion with Cross-Modal Attention:** Integrate an audio feature extraction branch (BPM, RMS energy, spectral brightness, major/minor mode) with the text embedding branch. Connect them via a cross-modal self-attention mechanism to ensure the engine gracefully processes instrumental tracks, EDM, and ambiguous rock.
- **Enrich Text Inputs with Interpretive Metadata:** Ingest crowd-sourced annotations and social tags alongside raw lyrics. This contextual data translates abstract artistic metaphors into standardized prose, significantly boosting downstream classification accuracy and alignment with human expectations.

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